

SEDALIA, MO, USA

WMO: 724453

Lat: 38.704N

Lon: 93.183W

Elev: 900

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	10.4	-3.4	4.7	7.5	0.8	5.9	12.4	27.0	39.3	24.8	37.9	7.5	320	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	96.5	75.8	92.8	75.8	90.3	75.6	79.6	90.2	78.2	88.8	77.0	87.2	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.6	143.6	86.3	75.1	136.2	84.8	73.5	129.0	82.7	43.8	90.0	42.3	89.1	41.0	87.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.9	20.5	18.6	-1.1	99.1	4.8	4.1	-4.5	102.0	-7.3	104.4	-10.0	106.7	-13.5	109.6		
			-1.9	81.8	4.6	1.2	-5.3	82.7	-8.0	83.4	-10.5	84.0	-13.9	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	31.9	35.8	46.3	56.5	66.0	74.7	78.8	77.4	69.9	57.9	46.5	35.4
	DBStd	18.81	11.73	11.90	11.25	9.39	7.97	6.09	5.70	5.76	7.19	9.45	10.38	11.02
	HDD50	1921	568	418	206	43	2	0	0	0	0	34	184	466
	HDD65	4578	1025	818	587	283	88	6	0	1	32	260	559	919
	CDD50	4300	7	20	90	237	499	742	892	848	597	277	78	13
	CDD65	1483	0	0	6	27	120	298	427	384	179	39	3	0
	CDH74	14363	0	2	40	220	959	2822	4612	3836	1565	292	15	0
	CDH80	5878	0	0	3	34	260	1094	2122	1730	568	67	0	0
Wind	WSAvg	9.1	10.2	10.2	10.6	10.6	9.0	8.4	7.0	6.8	7.4	8.8	9.8	9.9
Precipitation	PrecAvg	42.2	1.7	1.9	2.8	4.6	5.6	4.9	4.4	3.3	4.3	3.4	3.1	2.1
	PrecMax	60.1	5.5	5.1	6.5	12.7	15.3	12.9	13.4	9.5	13.3	10.0	10.2	5.6
	PrecMin	26.7	0.3	0.0	1.0	1.2	1.1	1.3	0.5	1.3	0.9	0.4	0.9	0.6
	PrecStd	8.3	1.1	1.3	1.4	1.9	3.1	2.7	3.3	1.8	2.9	2.1	2.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	72.2	78.8	83.8	89.3	95.4	101.4	100.2	93.5	86.0	76.1	67.7
		MCWB	53.8	58.2	63.3	67.1	72.6	76.8	75.7	74.7	73.9	71.0	62.0	59.8
	2%	DB	60.1	64.9	73.5	80.2	85.8	91.8	96.9	96.4	90.0	81.5	71.5	62.6
		MCWB	51.4	54.3	60.1	64.3	71.1	75.8	76.5	75.1	73.7	67.4	59.6	55.8
	5%	DB	54.4	60.1	69.4	75.9	82.8	89.6	93.1	92.5	86.3	76.8	66.3	56.1
		MCWB	46.1	51.5	57.9	62.0	70.0	75.3	76.7	75.1	72.2	64.6	57.2	49.1
	10%	DB	48.3	54.0	64.1	72.4	79.8	86.4	90.3	89.2	82.5	72.9	62.3	51.7
		MCWB	42.0	46.6	54.8	60.4	68.0	73.8	76.7	75.1	70.2	63.6	54.3	45.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	61.0	65.6	69.3	75.8	79.9	81.5	80.8	77.5	73.4	65.2	61.8
		MCDB	63.7	68.5	76.1	81.0	85.8	90.4	91.8	91.2	88.6	82.2	72.4	67.5
	2%	WB	52.4	56.9	62.5	66.8	73.6	78.1	80.0	78.9	75.4	70.6	61.6	56.6
		MCDB	57.7	63.2	71.0	76.4	83.2	88.3	90.7	89.6	85.6	77.4	68.0	61.5
	5%	WB	46.8	51.9	59.7	64.4	71.4	76.6	78.7	77.4	74.1	67.7	58.7	50.9
		MCDB	53.3	59.1	67.2	73.2	80.2	86.4	89.3	88.1	83.3	73.8	65.0	55.0
	10%	WB	42.4	46.2	55.7	62.1	69.5	75.1	77.6	75.9	72.4	64.9	55.4	46.1
		MCDB	48.3	53.5	63.0	70.4	77.1	84.4	88.0	85.9	80.4	71.4	61.3	50.4
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.4	18.9	19.7	18.7	18.2	18.5	19.6	20.5	19.4	18.1	15.7
		MCDBR	24.6	25.5	24.2	24.0	21.5	20.5	21.3	23.4	22.9	22.8	23.0	22.0
		MCWBR	17.3	17.4	14.1	13.6	10.9	9.2	7.6	8.2	9.9	11.6	14.8	16.0
	5% WB	MCDBR	22.6	24.3	20.8	20.4	19.3	18.4	18.5	19.6	19.7	19.0	20.3	20.6
		MCWBR	17.3	18.2	13.7	13.2	11.0	9.5	8.6	8.9	9.5	11.9	15.2	16.8
Clear-Sky Solar Irradiance	taub	0.297	0.311	0.337	0.388	0.424	0.427	0.446	0.450	0.387	0.345	0.307	0.293	
	taud	2.466	2.427	2.410	2.279	2.241	2.266	2.226	2.223	2.361	2.466	2.517	2.502	
	Ebn at Noon	280	291	293	282	272	271	265	260	272	275	275	274	
	Edh at Noon	25	30	34	41	44	43	44	43	36	29	24	23	
All-Sky Solar Radiation	RadAvg	685	914	1206	1561	1800	2037	2027	1784	1528	1085	747	589	
	RadStd	79	104	109	102	160	168	128	101	99	115	78	58	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page